Laboratory Protocol

Lab 1: Axial Tensile Testing and Strain Measurement Accuracy

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Course / Programme: TSM2370 Materialteknikk

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Institution / Lab: University of South-Eastern Norway

Experiment Date: [DD/MM/YYYY]

Lab Partner(s): [Lab Partner Names]

1. Objective

Determine how an aluminum alloy (6061-T6 or 5052-H32) behaves under a controlled, axial pulling load until fracture. The experiment will also demonstrate the critical importance of using an extensometer compared to machine crosshead travel for accurate strain calculation.

2. Theoretical Background

Engineering stress (σ) is the applied axial force (F) divided by the original cross-sectional area (A_0). Engineering strain (ε) is the change in length (ΔL) divided by the original gauge length (L_0). While the UTM load cell accurately measures force, calculating strain using crosshead travel introduces errors. An extensometer is required to directly measure elongation over the gauge length. The material progresses from linear elastic deformation into plastic deformation, characterized by strain hardening, necking, and fracture.

2.1 Relevant Formulas

Engineering Stress: $\sigma = F / A_o$

Engineering Strain: $\varepsilon = \Delta L / L_o$

Young's Modulus: $E = \sigma / \varepsilon$

3. Materials & Equipment

Device / Chemical	Amount / Concentration	Manufacturer	Notes
Universal Testing Machine	10 kN (Max 5kN chucks)	Lloyd (LR10K)	Used for axial pulling
Camera Extensometer	1 unit	Basler/Logitech	Optical strain measurement
Aluminum Sheet	1 mm thick, 12.5 mm	ISO Standard	6061-T6 or 5052-H32

Dog-bones	width		
Digital Calipers	1 unit	-	For L0 measurement

4. Experimental Procedure

4.1 Experimental Setup

The Lloyd LR10K is equipped with tensile grips. The camera extensometer is mounted facing the specimen to track optical strain markers on the gauge section. The aluminum dog-bone is centered within the grips.

4.2 Procedure Steps

1. Take safety precautions and ensure the UTM safety shields are in place.
2. Use digital calipers to measure the exact initial gauge length (L0) and width/thickness of the stamped aluminum part.
3. Input the measured L0 into the extensometer software and clamp the specimen securely in the UTM grips.
4. Initiate the tensile pull at the specified crosshead speed.
5. Run the test until complete specimen fracture.
6. Export the resulting raw force and elongation data from the LR10K software.
7. Clean the workstation and safely dispose of the fractured aluminum pieces.

5. Results & Measurements

Parameter / Measurement	Value	Unit	Note
Yield Strength (Machine Travel)		MPa	
Yield Strength (Extensometer)		MPa	
Ultimate Tensile Strength (UTS)		MPa	
Young's Modulus (Machine Travel)		GPa	
Young's Modulus (Extensometer)		GPa	

5.1 Calculations

Calculate the Yield Strength, Ultimate Tensile Strength (UTS), and Young's Modulus (E) for both datasets (machine travel vs. extensometer).

5.2 Graphs / Diagrams

[Insert diagram or figure here (don't forget labels and units). Plotting can be done using Python or Excel.]

6. Discussion

6.1 Interpretation of Results

Discuss the variance in the linear elastic slope (E) between the two curves. Explain why the extensometer is mandatory for extracting true material properties.

6.2 Error Analysis

Systematic errors:
[Error source 1]

Random errors:
[Error source 1]

7. Conclusion

[Summarise the most important findings and answer the question posed at the beginning]

8. References

[1] Häll, K. (2026). Materials Science & Engineering: Practical Characterization (Practical_Materials_Characterization_tests.pdf). University of South-Eastern Norway.

[2] USN Department of Science and Industry Systems (2026). TSM2370 Materialteknikk 2026 Course Material (TSM2370 Materialteknikk 2026). University of South-Eastern Norway.

9. Appendix

9.1 Raw Data

[Raw data tables or measurement sheets]

9.2 Additional Figures

[Further graphs, photos, or sketches]

Date: _____ Signature: _____

I hereby confirm that this protocol was written independently and truthfully.

Laboratory Protocol

Lab 2: Thermal Processing and Quench Severity

Author: [First Last Name]

Student ID: [Student ID]

Course / Programme: TSM2370 Materialteknikk

Supervisor: Rasmus Ullestad Relling & Kent Häll

Institution / Lab: University of South-Eastern Norway

Experiment Date: [DD/MM/YYYY]

Lab Partner(s): [Lab Partner Names]

1. Objective

Measure localized resistance to plastic deformation across different microstructures of AISI 1045 steel. The experiment demonstrates how varying quench media (water vs. oil/hot water) and tempering processes dictate final mechanical properties.

2. Theoretical Background

Hardness correlates strongly with a material's yield strength. Rapidly quenching steel in water prevents carbon diffusion, forcing the FCC austenite into a BCT lattice (martensite) – the hardest and most brittle structure. A faster quench (cold water) is more severe than a slower quench (hot water/oil), leading to different hardness levels. Reheating (tempering) the as-quenched martensite relieves internal stresses and allows carbon to precipitate into cementite particles, restoring toughness.

2.1 Relevant Formulas

Vickers Hardness (HV): Calculated via indentation diagonal lengths (d1, d2) and applied test force.

3. Materials & Equipment

Device / Chemical	Amount / Concentration	Manufacturer	Notes
Medium-carbon steel blocks	15-25 mm width, 10-15 mm thick	AISI 1045	4 samples per group
Lab Furnaces	Capable of 1000°C	Tinius Olsen	Austenitizing and tempering
Quench Media Stations	20°C water, 70°C water, oil	-	For varying quench severity
Vickers Hardness	1 unit	-	For measuring

Tester			plastic deformation resistance
Polishing Equipment	As needed	-	To prepare surfaces for indentation

4. Experimental Procedure

4.1 Experimental Setup

Four steel samples (A, B, C, D) are prepared. Furnaces are preheated to 1000°C for austenitizing, and quench baths are brought to the designated temperatures. A lower-temperature furnace is prepared for tempering.

4.2 Procedure Steps

8. Take safety precautions and put on heat-resistant protective equipment (gloves, tongs, face shields).
9. Student 1 (Sample A): Polish the untreated 1045 steel block and measure the baseline Vickers hardness.
10. Student 2 (Sample B): Austenitize sample at 1000°C, hold for uniformity, then rapidly quench in 20°C cold water. Polish and measure hardness.
11. Student 3 (Sample C): Austenitize sample at 1000°C, then quench in 70°C hot water OR quenching oil. Polish and measure hardness.
12. Student 4 (Sample D): Austenitize at 1000°C and quench in 20°C water. Immediately transfer to the tempering furnace for the prescribed cycle. Air cool, polish, and measure hardness.
13. Record all hardness indentations and clear the testing equipment.

5. Results & Measurements

Parameter / Measurement	Value	Unit	Note
Sample A Hardness (Baseline)		HV	
Sample B Hardness (20°C Water Quench)		HV	
Sample C Hardness (Reduced Severity Quench)		HV	
Sample D Hardness (Tempered Martensite)		HV	

5.1 Calculations

Record the Vickers hardness values for all four samples in the table. Calculate average hardness if multiple indentations are performed per sample.

5.2 Graphs / Diagrams

[Insert diagram or figure here (don't forget labels and units). Plotting can be done using Python or Excel.]

6. Discussion

6.1 Interpretation of Results

Compare Sample B and Sample C to explain the concept of quench severity and cooling rates. Compare Sample B and Sample D to explain the microstructural effects of tempering, the relief of internal stresses, and the restoration of toughness.

6.2 Error Analysis

Systematic errors:
[Error source 1]

Random errors:
[Error source 1]

7. Conclusion

[Summarise the most important findings and answer the question posed at the beginning]

8. References

[1] Häll, K. (2026). Materials Science & Engineering: Practical Characterization (Practical_Materials_Characterization_tests.pdf). University of South-Eastern Norway.

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